

Mathematics Teaching Reference Handbook

Linear equations

An equation states that two expressions are equal. Solving means finding values that keep the equality true. When the same operation is applied to both sides, equality is preserved.

One-step equations isolate the variable with one inverse operation. Two-step equations require learners to decide the order of undoing operations. Checking by substitution should be included so the process does not become mechanical.

Common mistakes include changing only one side, losing a negative sign, and applying an operation to one term instead of an entire side.

Graphs and slope

A coordinate graph connects numeric relationships to position. Slope measures vertical change divided by horizontal change. Learners should calculate slope from two points and interpret what the sign means.

A positive slope rises from left to right, a negative slope falls, a zero slope is horizontal, and a vertical line has undefined slope. The y-intercept is where x is zero.

Graph questions should include interpretation, not only calculation. For example, a slope of 3 in a savings graph may mean an increase of three units per time step.

Quadratics

A quadratic expression contains a squared variable term. Learners may expand simple binomials, factor common patterns, and connect factors to zeros.

The discriminant can be introduced as a way to anticipate the number of real roots without requiring a full proof. The emphasis should remain on what the value tells us.

Graph shape and algebra should be linked carefully. A parabola opening upward does not by itself tell the learner whether there are zero, one, or two x-intercepts.

Data and statistics

Mean uses every value and is sensitive to outliers. Median is the middle ordered value and may better represent a skewed dataset. Mode is the most frequent value and may be absent or have multiple values.

Charts should be read critically. A truncated axis can exaggerate a small difference. Percent change and percentage points are not the same thing.

Activities should use small datasets that learners can check by hand before moving to larger tables.